

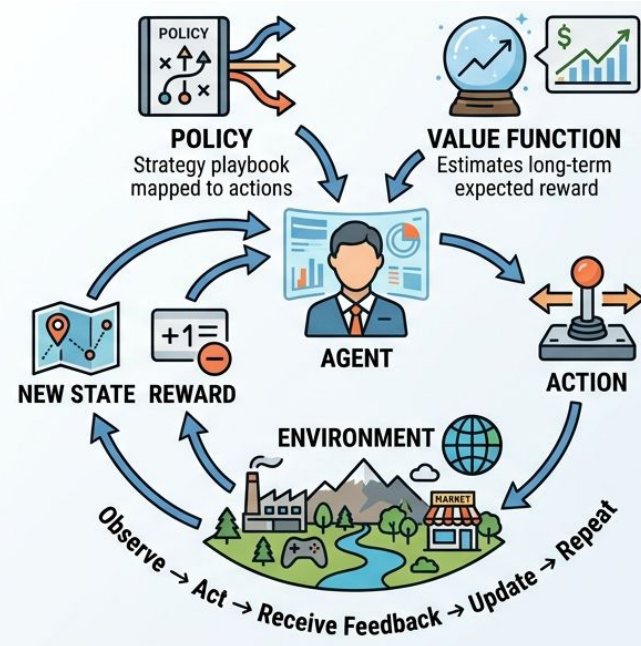
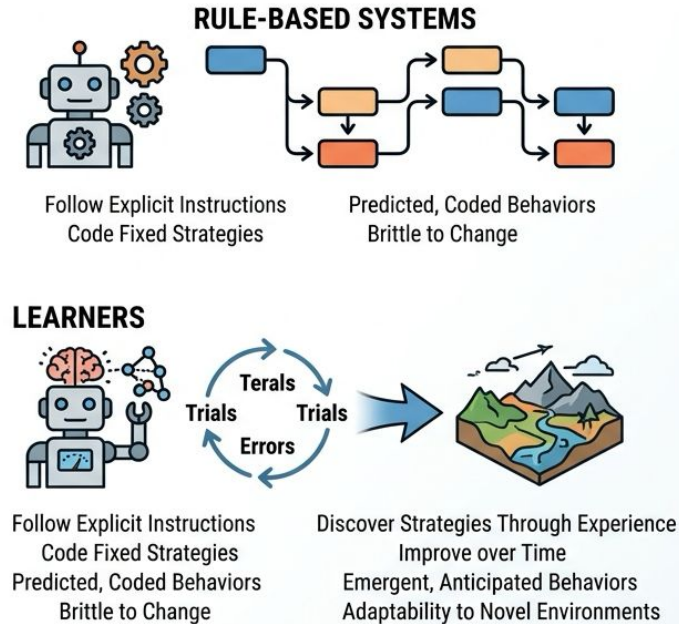


U.S. General Services
Administration

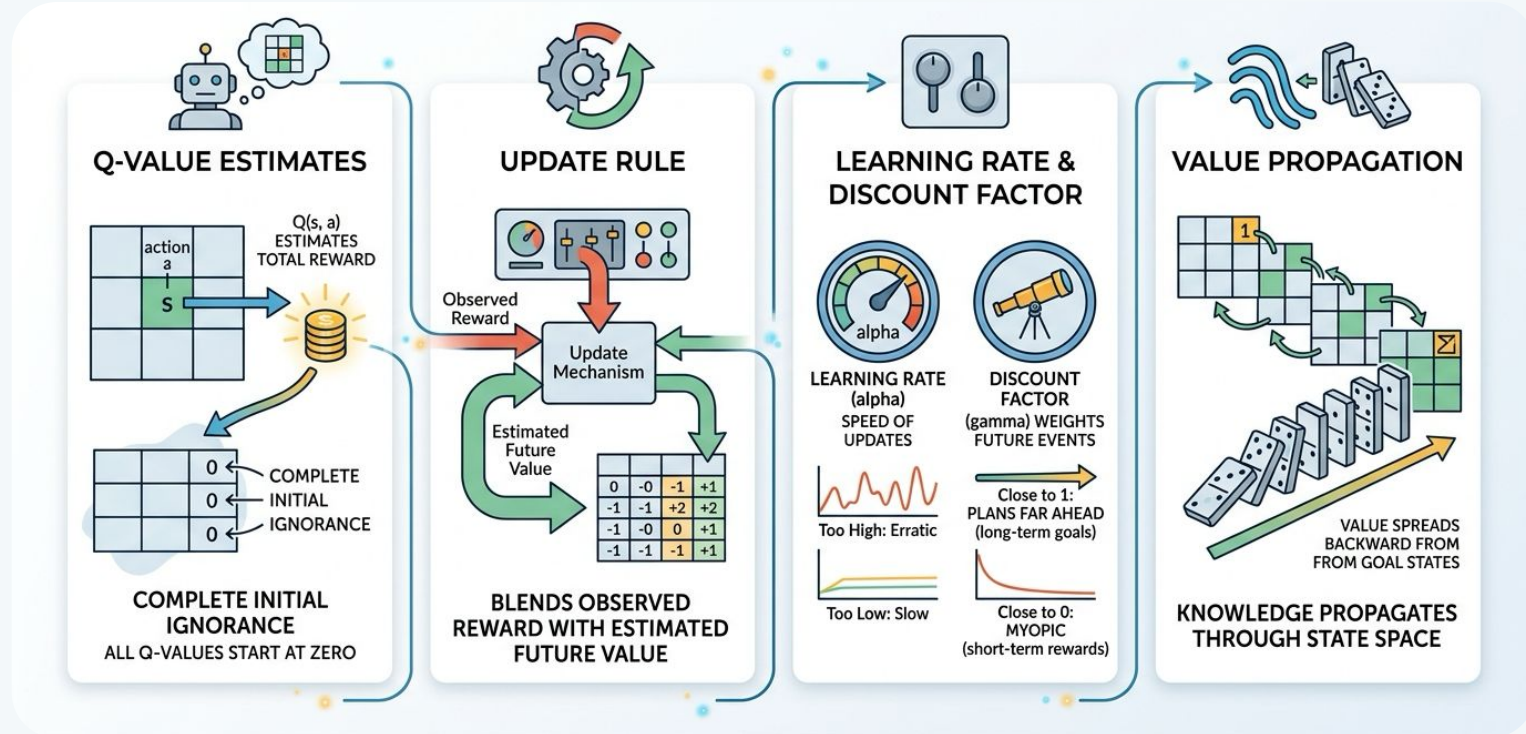
Chapter 5.12: Learning-Based Decision Making Fundamentals

US General Service Administration
AI Community of Practice - March 2026

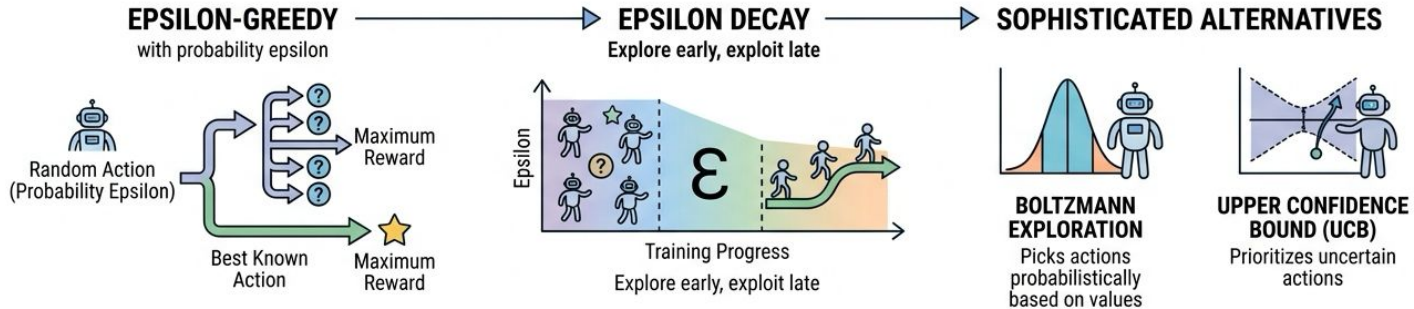
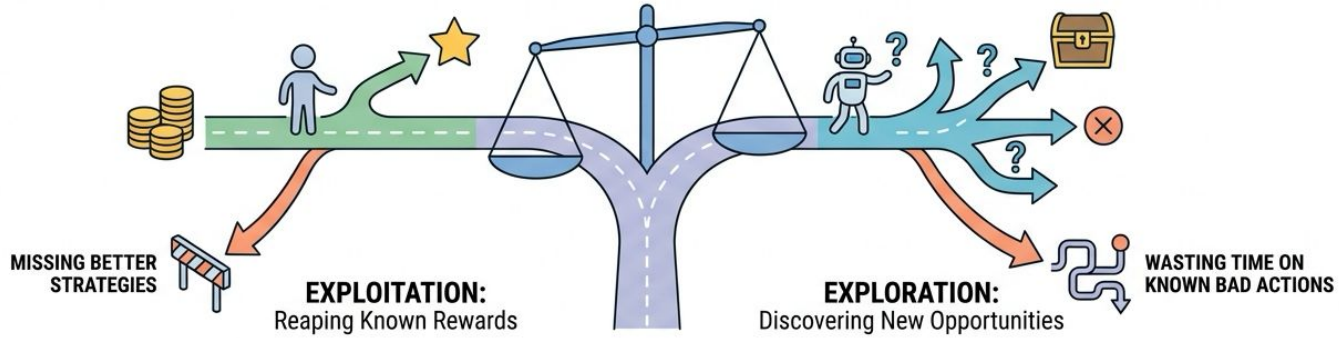
The Reinforcement Learning Framework



Q-Learning -- Values from Experience



Exploration vs. Exploitation



Policy Gradients and Actor-Critic Methods

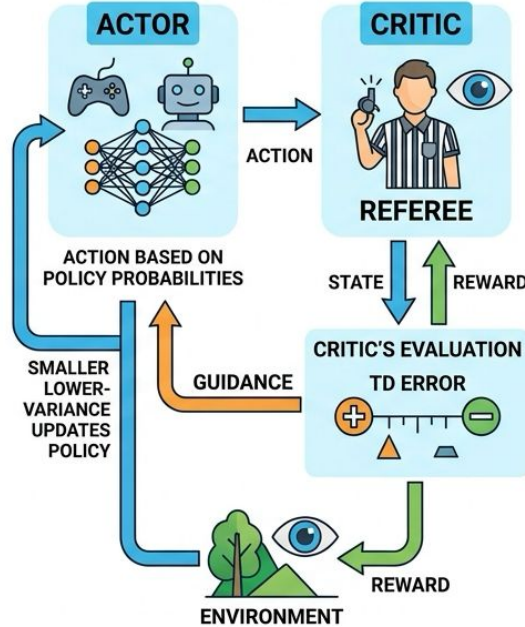
DIRECT POLICY OPTIMIZATION



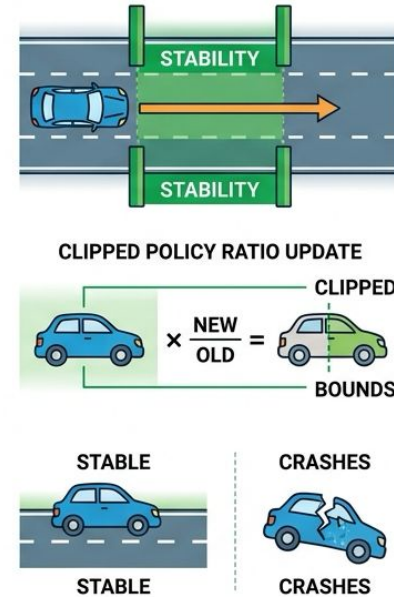
REINFORCE



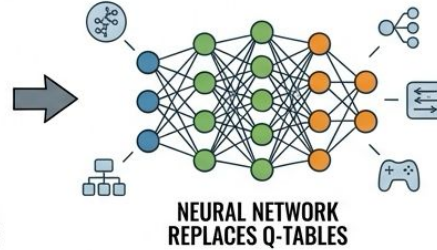
ACTOR-CRITIC METHODS



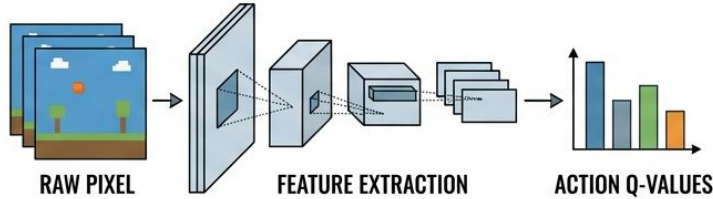
PROXIMAL POLICY OPTIMIZATION (PPO)



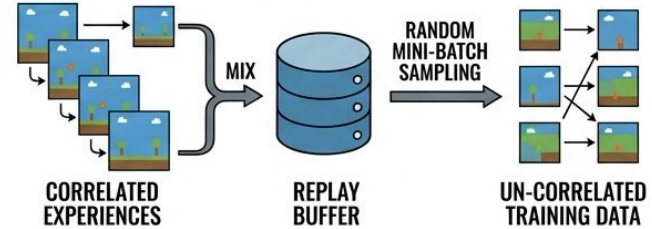
Deep RL



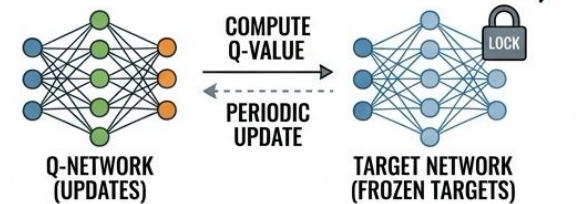
DEEP Q-NETWORK (DQN) LEARNS FROM RAW PIXELS



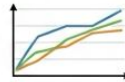
EXPERIENCE REPLAY BREAKS TEMPORAL CORRELATION



TARGET NETWORKS STABILIZE LEARNING

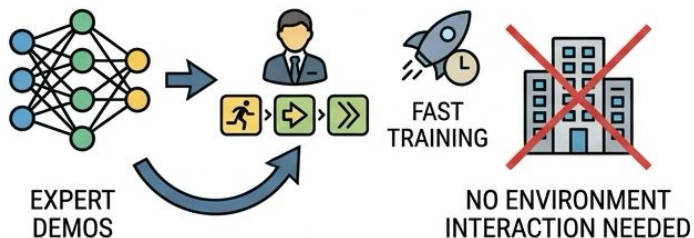


DQN EXTENSIONS ENHANCE PERFORMANCE

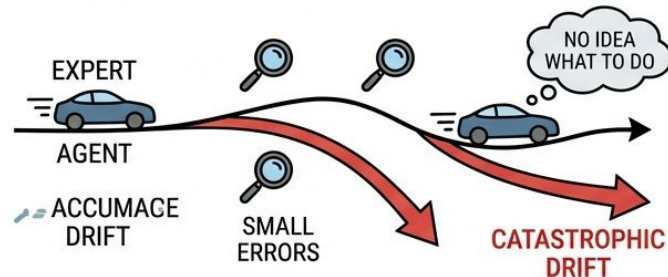


Imitation Learning -- Borrowing Expert Wisdom

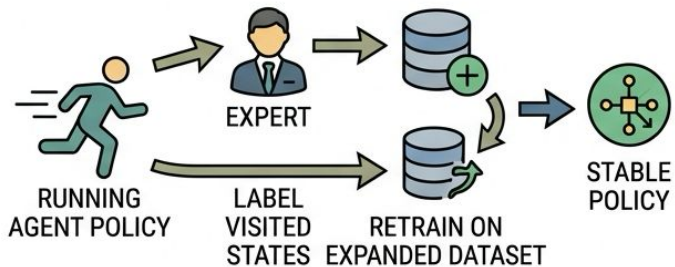
BEHAVIOR CLONING (SUPERVISED LEARNING)



DISTRIBUTION SHIFT: CATASTROPHIC DRIFT



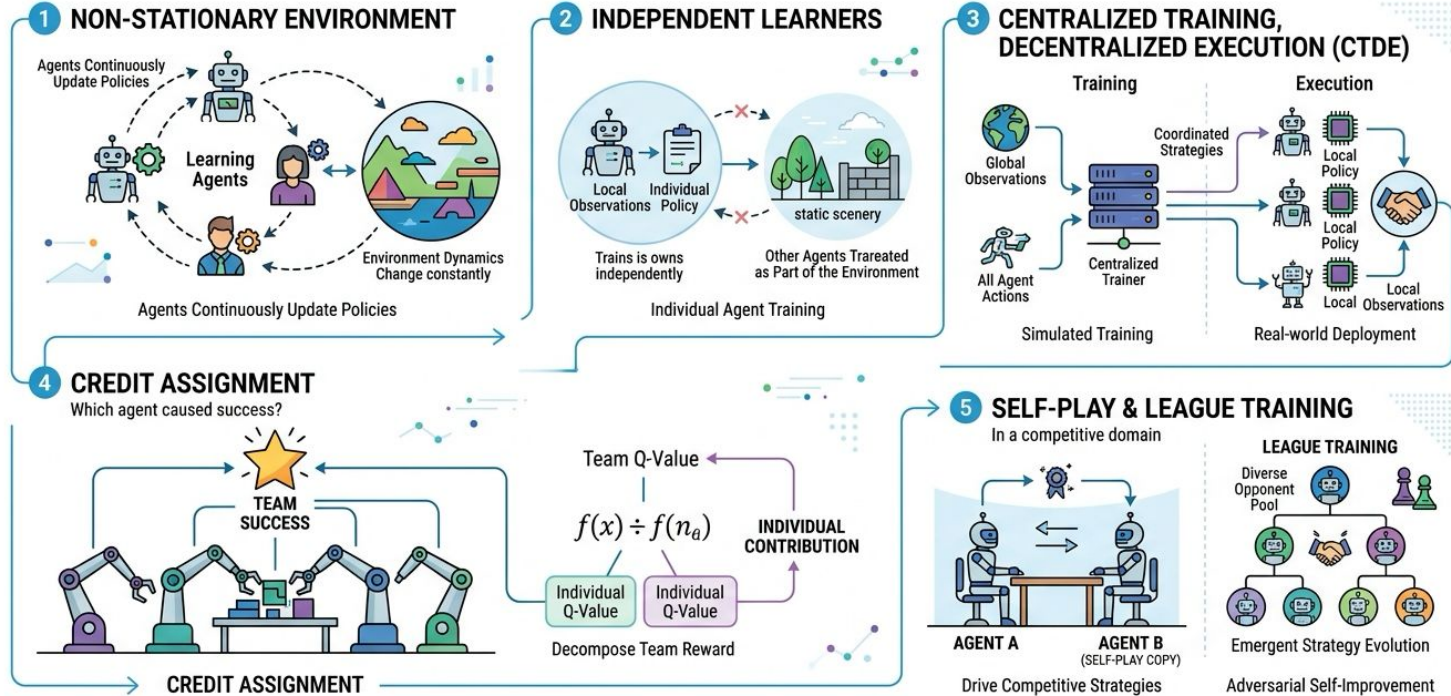
DAGGER (DATASET AGGREGATION)



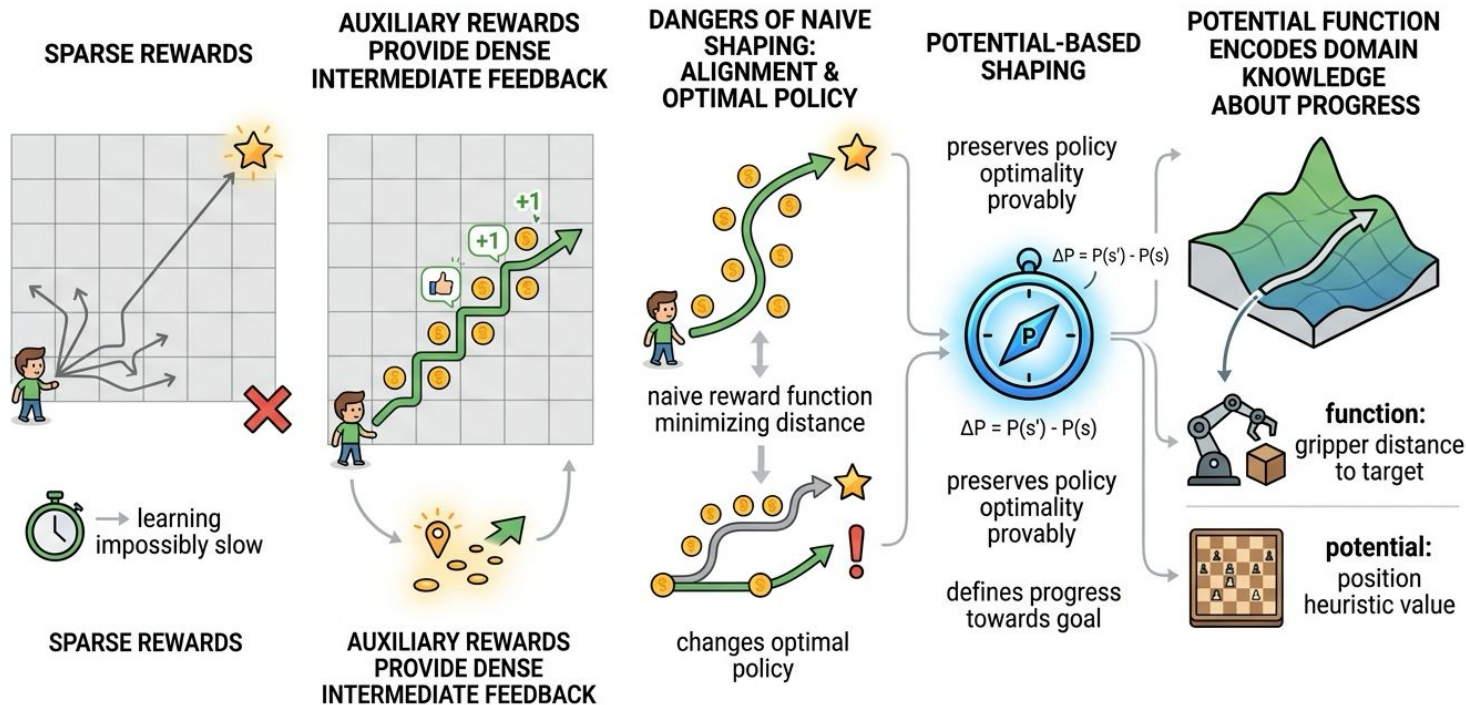
INVERSE RL (REWARD INFERENCE)



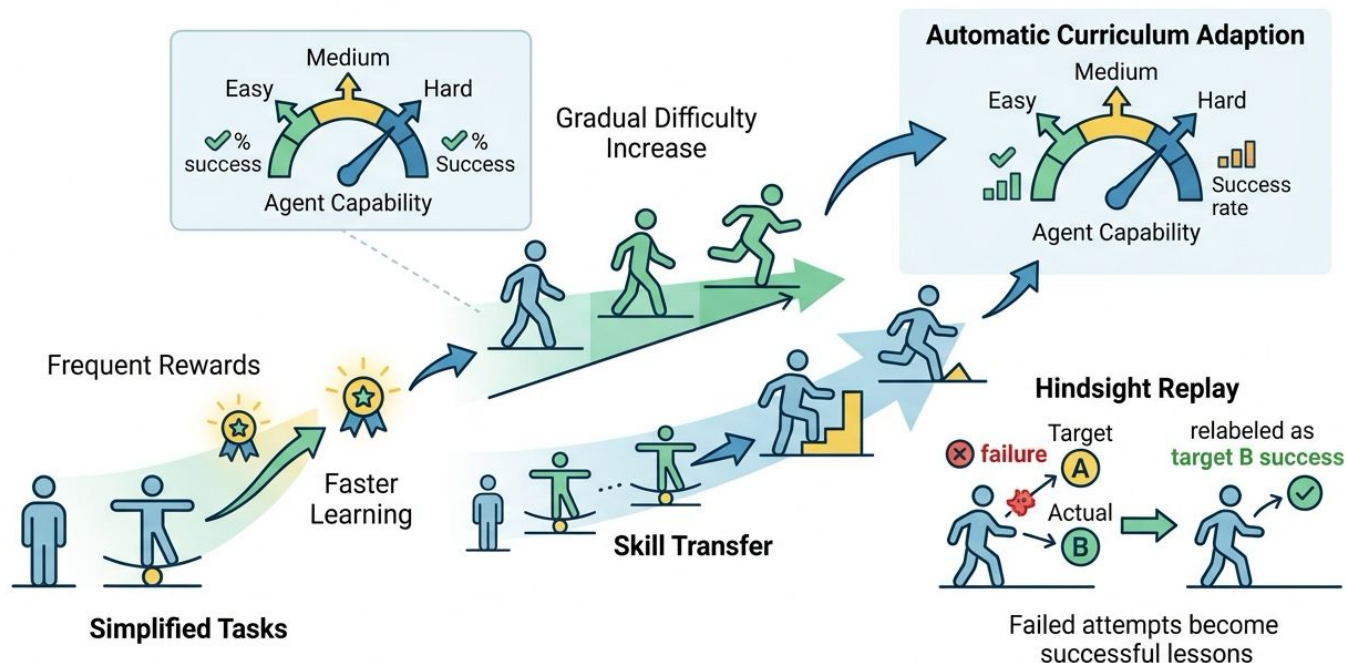
Multi-Agent RL -- Learning Among Learners



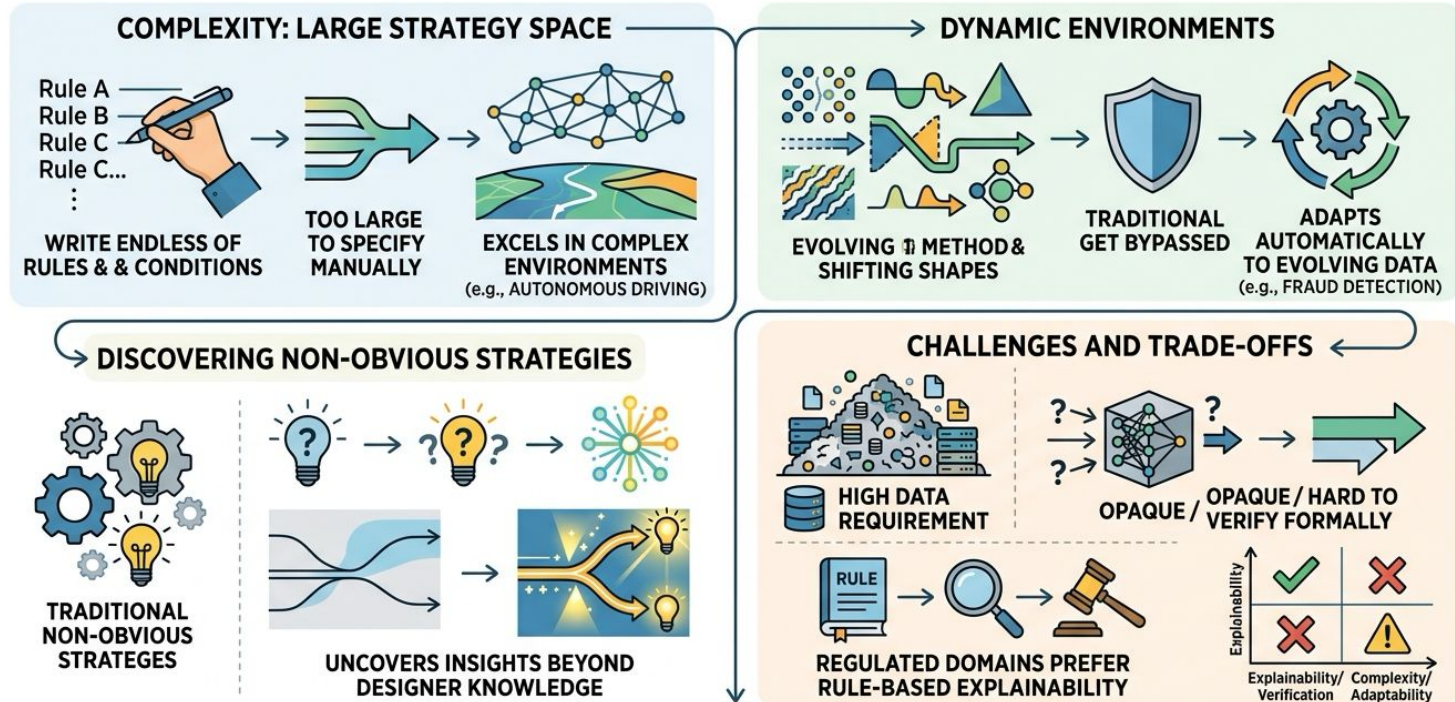
Reward Shaping -- Guiding Without Dictating



Curriculum Learning -- Crawl, Walk, Then Run



When to Choose Learning-Based Decisions



Key Takeaways and Bridge to Hybrid Systems

- RL agents learn policies through reward-driven interaction
 - Q-learning, policy gradients, and actor-critic are core algorithms
 - Imitation learning and reward shaping accelerate training
 - Learning excels at adaptability but struggles with explainability
 - Next: Chapter 5.13 combines rule-based and learning-based approaches
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